

Multi-condition differential cell-cell communication inference using Spatial Transcriptomics Data

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Abstract

Cell communication shapes the tumour microenvironment, but current computational models often ignore spatial proximity and condition specific signals. This gap limits the ability to capture realistic changes in interactions between normal and cancer tissue. Our aim is to build a unified framework that measures communication differences while also accounting for spatial context and latent biological shifts. We combine spatial interaction matrices from COMMOT and adapt the SCDISInFact framework, which was originally designed for single-cell gene expression disentanglement, and repurpose it to operate on ligand–receptor features. This unconventional use allows the model to learn shared and condition-driven factors, generate normal-like counterfactual profiles, and quantify changes in communication strength. We then group interaction pairs into biologically interpretable programs and analyse changes across networks and cell types. The results show strong cancer related increases in invasion, extracellular matrix activity, angiogenesis and immune signalling, while the spatial model removes long range artefacts and uncovers local communication hotspots. This integrated approach provides a coherent and spatially grounded view of communication rewiring in cancer and offers a general method for spatial transcriptomics analysis.

Keywords

Ligand, Receptor, Cell-cell Communication

1 Introduction

Cell–cell communication (CCC) is central to tissue organisation, immune surveillance, angiogenesis, and tumour progression [1]. These interactions are orchestrated through ligand–receptor (LR) signalling networks that define how cells exchange biochemical information. Recent computational frameworks such as

CellChat, CellPhoneDB, and COMMOT have substantially improved the ability to infer CCC from transcriptomic data [2, 3, 4]. However, despite this progress, two major gaps remain. First, most CCC methods neglect spatial context by assuming that any cell can signal to any other cell. This assumption contradicts physical diffusion limits and receptor accessibility, and mounting evidence from spatial transcriptomics shows that spatial structure can drastically reshape inferred signalling patterns [5, 6]. Second, current differential CCC pipelines are limited in capturing non-linear, condition-specific biology [7]. Changes in LR signalling between normal and cancer tissues often arise from complex latent programs, yet most approaches rely on simple effect-size comparisons. SCDISInFact offers a principled way to disentangle shared, batch, and condition-driven variation, but it has not previously been applied to LR features [8].

These limitations motivate a unified framework capable of quantifying LR communication at spot-level resolution, integrating spatial proximity and neighbourhood structure, and uncovering latent LR programs that drive cancer progression. Understanding how cancer reshapes LR networks, and how these changes differ when spatial constraints are explicitly modelled, remains a fundamental and unresolved problem in tumour microenvironment (TME) research [9].

In this work, we address the overarching question: *How does cancer rewire ligand–receptor signalling programs, and to what extent are these changes dependent on spatial organisation within the tissue?* To this end, we integrate COMMOT-derived spatial interaction matrices with a repurposed version of the SCDISInFact model, which we adapt beyond its original gene-expression context to operate directly on LR interaction profiles. This enables us to learn shared and condition-driven latent factors, generate normal-like counterfactual predictions for cancer interactions, and quantify differential signalling across LR pairs, cell types, and spatial neighbourhoods. We further classify LR pairs into biologically interpretable programs; im-

immune, angiogenic, stromal/ECM, growth/survival, and EMT/invasion, to link latent factors to mechanism.

2 Materials & Methods

2.1 Data acquisition and preprocessing

Matched normal and cancer spatial transcriptomics sections were analysed using COMMOT-derived ligand–receptor (LR) communication matrices. Each dataset was stored as an `AnnData` object containing gene expression, inferred cell-type labels, and (x, y) tissue coordinates. For every curated LR pair, COMMOT provides a directed spot-by-spot adjacency matrix whose entries denote communication strength from a sender spot to a receiver spot. To obtain an edge-level representation suitable for statistical testing and latent factor modelling, each adjacency matrix was reshaped into a long-form table in which each row corresponds to a single sender→receiver communication event. The table includes the Ensembl IDs of the ligand and receptor, sender and receiver spot barcodes, their associated cell-type annotations, the LR communication score, and the spatial coordinates of both spots. Zero-valued entries were discarded so that downstream analyses focused on biologically active communication. The same procedure was applied independently to the normal and cancer sections.

2.2 Harmonisation of normal and cancer communication edges

To enable a direct condition-matched comparison, sender and receiver spot identifiers were cleaned to remove section-specific prefixes and ensure uniform nomenclature. Only those ligand–receptor pairs and spot-level edges that were detectable in both conditions were retained. The resulting harmonised normal/cancer edge tables contained matched communication events with identical sender–receiver coordinates and identical LR identities, permitting direct quantification of cancer-driven changes.

2.3 Differential ligand–receptor testing using the Mann–Whitney U test

Differential LR signalling between normal and cancer tissues was quantified using a non-parametric framework. For every LR pair and every sender–receiver cell-type combination, communication scores were aggregated across all spatially observed edges in normal and cancer. LR–cell-type combinations with fewer than three edges per condition were excluded to maintain statistical robustness.

Let $\{x_i^{(c)}\}$ and $\{x_j^{(n)}\}$ denote the sets of COMMOT LR scores observed in cancer and normal tissues, respectively. Because LR scores exhibit non-Gaussian

distributions and may contain outliers, the central tendency was summarised using the median:

$$\tilde{x}^{(c)} = \text{median}\left(x_i^{(c)}\right), \quad \tilde{x}^{(n)} = \text{median}\left(x_j^{(n)}\right).$$

A $\log_2$ fold-change was computed as

$$\log_2 \text{FC} = \log_2 \left(\frac{\tilde{x}^{(c)} + 10^{-6}}{\tilde{x}^{(n)} + 10^{-6}} \right),$$

where the constant prevents division by zero. Statistical significance was evaluated using the two-sided Mann–Whitney U test [10]:

$$p = \text{MWU}\left(\{x_i^{(c)}\}, \{x_j^{(n)}\}\right).$$

A ligand–receptor pair was defined as strongly differential when $|\log_2 \text{FC}| > 1$ and $p < 0.05$. Significant interactions were annotated with gene symbols using `MyGene.info` and visualised as a directed ligand→receptor network using `networkx`, with edge width proportional to the magnitude of $\log_2 \text{FC}$.

2.4 Adaptation of SCDISInFact for ligand–receptor communication modelling

SCDISInFact, originally developed for disentangling gene-expression variation, was repurposed to operate on LR communication vectors. Each spot–spot edge was encoded as a feature vector containing all LR communication scores associated with that edge. Metadata describing condition (normal or cancer), inferred batch, sender and receiver cell types, and spatial descriptors (described below) were included as covariates. SCDISInFact was trained to infer a shared latent representation capturing baseline communication structure, a condition-specific latent factor capturing cancer-associated perturbations, and a batch-associated factor. (Figure 1)

To isolate biological changes attributable to cancer, a *normal-counterfactual* was generated for every cancer edge by decoding its latent representation while setting the condition code to the normal state. This yielded an estimate of what each cancer communication event would look like under normal physiology [11]. The model-based differential communication score was defined as

$$\Delta LR = LR_{\text{observed cancer}} - LR_{\text{SCDISInFact-normal}}.$$

The SCDISInFact model was run in two variants: (i) a non-spatial setting that utilised only LR features, and (ii) a spatially informed setting that integrated distance and neighbourhood descriptors. This allowed the contribution of spatial constraints to be separately quantified.

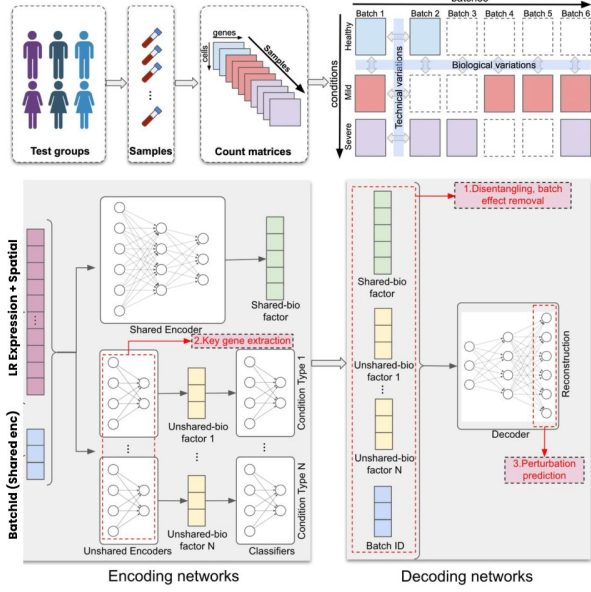


Figure 1: Repurposed scDisInFact Architecture for integrating spatial features to aid differential cell-cell communication

2.5 Construction of ligand–receptor signalling programs

To enable mechanistic interpretation, LR pairs were assigned to one of six functional programs: immune, angiogenic, stromal/ECM, growth/survival, EMT/invasion, or “other”. This classification was curated using established literature describing pathway-specific ligand–receptor axes, including integrin-binding ECM ligands, PDGF/VEGF angiogenic factors, TGF β - and SEMA/PLXN-associated EMT drivers, and immune-modulatory interactions such as IL16–CD4 and LGALS9–CD44. This mapping enabled aggregation of differential signals into biologically coherent modules.

2.6 Spatial feature engineering

To incorporate spatial context into the SCDISInFact model, each LR edge was augmented with microenvironmental descriptors derived from the spot coordinates. The Euclidean distance between sender and receiver spots was computed for every edge, and both the raw distance and its logarithmic transform were included as features. To capture local signalling niches, a neighbourhood-based hotspot score was computed by averaging LR activity across the k nearest neighbours of each spot ($k = 10$). Each edge inherited the hotspot values of its sender and receiver, providing a coarse-grained summary of its surrounding microenvironment. These spatial descriptors were concatenated with the high-dimensional LR vectors prior to SCDISInFact training.

2.7 Program-level aggregation and network construction

Differential communication scores derived from SCDISInFact were aggregated at the level of signalling programs by averaging ΔLR across all LR pairs assigned to each program. This yielded sender→receiver program activity matrices that revealed how communication between specific cell types was reorganised in cancer. These matrices were visualised using heatmaps. At the interaction level, the most strongly differential LR pairs were identified and assembled into directed ligand→receptor networks constructed with **networkx**. Edge weights reflected the magnitude of communication gain in cancer, whereas node colours encoded ligand or receptor identity. This network representation revealed modular organisation of angiogenic, stromal/EMT, and immune-regulatory signalling pathways and enabled direct comparison between spatial and non-spatial SCDISInFact variants.

3 Results

3.1 Differential Cell–Cell Communication Identified by Mann–Whitney Testing

To quantify differences in ligand–receptor (LR) signalling between normal and cancer tissue, we performed a Mann–Whitney U test across all LR–cell-type combinations shared between conditions. We considered interactions significant if $|\log_2 FC| > 1$ and $p < 0.05$.

Overview of Significant Interactions. Several LR interactions exhibited strong differential activity (Table 1). A subset of immune-regulatory interactions was markedly *downregulated* in cancer, including **GRN→SORT1** ($\log_2 FC = -4.77$) and **IL16→CD4** ($\log_2 FC = -4.83$), pointing to suppressed immune homeostasis in the tumour microenvironment. Conversely, growth-factor and EMT-associated pathways were enriched in cancer: **PGF→FLT1**, **PDGFC→PDGFRA**, **ANGPTL4→CDH11/SDC3**, and **GAS6/PROS1→AXL** all showed substantial upregulation.

Network-Level Interpretation. The differential network (Figure 2) formed distinct modules: (1) A PDGF/VEGF angiogenic cluster, (2) An AXL-driven EMT module, and (3) A suppressed immune-signalling module (CD4, CD44, SORT1). This modular organisation is consistent with canonical tumour progression pathways—enhanced angiogenesis, stromal remodelling, and EMT accompanied by reduced immune regulatory signalling.

Table 1: Significant LR interactions from Mann–Whitney U testing between cancer and normal (subset).

log2FC	pvalue	n_edges_cancer	n_edges_normal	direction	ligand_symbol	receptor_symbol
-4.770037	0.037926	222	142	Up in Normal	GRN	SORT1
1.143008	0.036589	18	20	Up in Cancer	PGF	FLT1
2.029497	0.000004	10	53	Up in Cancer	PDGFC	PDGFRA
-1.944358	0.017570	175	53	Up in Normal	IL34	CSF1R
1.250857	0.018211	8	84	Up in Cancer	ANGPTL4	CDH11
2.114535	0.035447	6	97	Up in Cancer	ANGPTL4	SDC3
-3.550658	0.000007	898	153	Up in Normal	LGALS9	CD44
-4.827635	0.038042	332	14	Up in Normal	IL16	CD4
1.166056	0.000119	147	314	Up in Cancer	GAS6	AXL
-1.565446	0.003049	211	65	Up in Normal	CSF1	CSF1R
1.342291	0.030963	15	33	Up in Cancer	PROS1	AXL
1.411631	0.013214	23	110	Up in Cancer	PDGFA	PDGFRB
1.696783	0.045465	10	70	Up in Cancer	PDGFA	PDGFRA
-5.090445	0.000248	343	105	Up in Normal	TNFSF12	TNFRSF12A

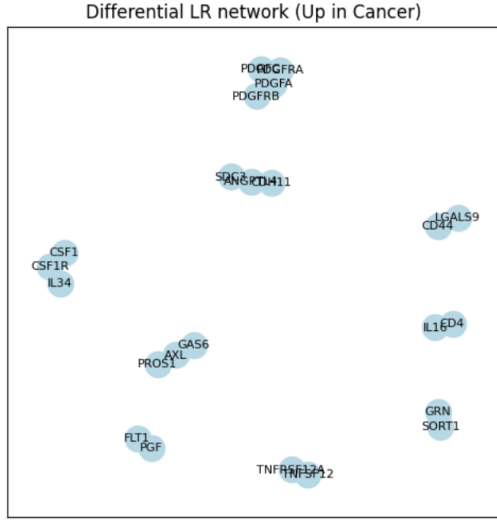


Figure 2: Differential LR network (Up in Cancer). Nodes represent ligands or receptors; edges denote significantly increased LR interactions in cancer.

Overall, the Mann–Whitney analysis highlights coherent rewiring of communication programs in cancer, reinforcing the biological plausibility of our LR-inference framework.

3.2 Non-Spatial SCDISInFact Analysis

Latent Factor Structure and LR Program Organisation Applying SCDISInFact to the non-spatial CCC matrices revealed coherent structure in the ligand–receptor landscape. Hierarchical clustering of LR pairs (Figure 3) demonstrated that interactions segregate into biologically recognisable modules, including EMT/invasion, stromal–ECM signalling, immune regulation, and growth-factor pathways. This provided empirical justification for aggregating LR pairs into mechanistic “programs” for downstream interpretation.

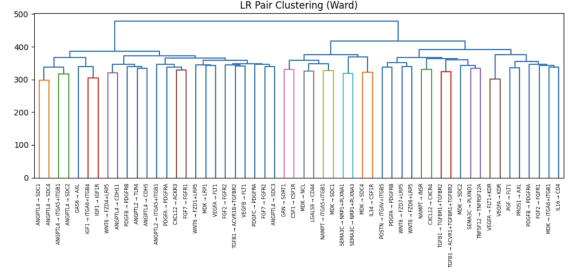


Figure 3: Hierarchical clustering of ligand–receptor pairs in the non-spatial SCDISInFact model.

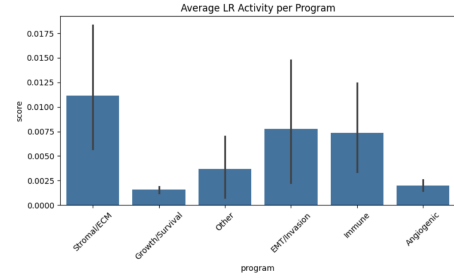


Figure 4: Mean absolute ligand–receptor activity per biological program (non-spatial SCDISInFact).

Differential Programme Activation in Cancer

To assess how cancer alters these signalling programs, we computed the mean absolute deviation in LR activity across all edges for each program (Figure 4). Stromal/ECM and EMT/invasion programs showed the largest increases in cancer, consistent with enhanced matrix remodelling and motility-associated signalling. Immune-related programs displayed moderate activation differences, whereas angiogenic and growth-factor programs contributed comparatively smaller shifts in the non-spatial setting. These rankings reflect the dominant, condition-specific axes of variation captured by SCDISInFact.

Cell-Type-Specific Organisation of LR Programs

To resolve how cancer reshapes communication between specific cell types, we evaluated program-level LR activity for each sender–receiver pair using non-spatial SCDISInFact. Three dominant patterns emerged.

First, Stromal/ECM signalling displayed the strongest and most widespread increases, with endothelial→endothelial, smooth muscle→smooth muscle edges showing pronounced elevation (Figure 5). This agrees with established roles of ECM remodelling, integrin signalling, and matrix–immune crosstalk in tumour progression.

Second, EMT/Invasion programs showed elevated activity across multiple epithelial-derived signalling routes, particularly smooth muscle→smooth muscle and epithelial→myeloid interactions (Figure 6). These patterns reflect canonical shifts in invasion-associated ligand–receptor axes (e.g., AXL,

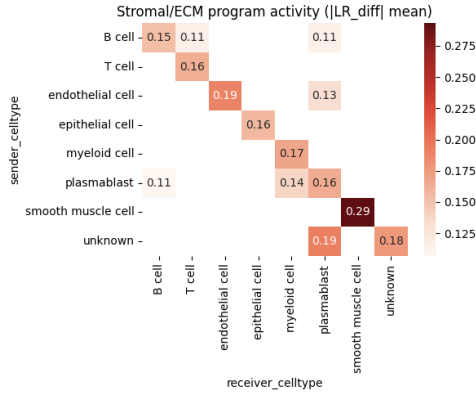


Figure 5: Program-level LR activity across sender-receiver cell-type pairs inferred using non-spatial SCDISInFact. Stromal/ECM, EMT/Invasion, and Immune programs show the strongest differential patterns.

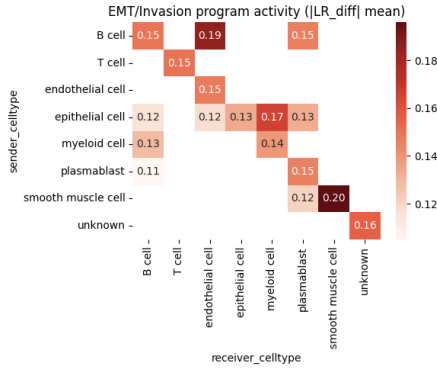


Figure 6: Program-level LR activity across sender-receiver cell-type pairs inferred using non-spatial SCDISInFact. Stromal/ECM, EMT/Invasion, and Immune programs show the strongest differential patterns.

SEMA/PLXN, and TGFB-related pathways).

Third, immune-related programs exhibited mixed behaviour, with increased signalling predominantly routed through myeloid and plasmablast populations (Figure 7). These patterns are consistent with immune-stromal interplay and partial immune suppression observed in tumours.

Differential Ligand-Receptor Interactions and Network Topology Using SCDISInFact without spatial information, we quantified cancer-specific changes in communication strength by contrasting observed cancer LR scores with model-predicted “normal-like” profiles. The strongest interactions, ranked by their maximal differential value, are summarised in Table 2. These high-confidence interactions reveal coherent biological themes.

First, we observed extensive upregulation of

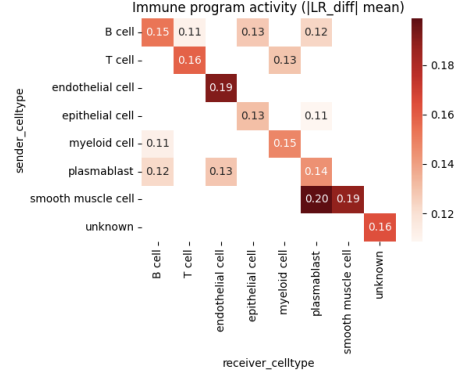


Figure 7: Program-level LR activity across sender-receiver cell-type pairs inferred using non-spatial SCDISInFact. Stromal/ECM, EMT/Invasion, and Immune programs show the strongest differential patterns.

midkine (MDK)-driven signalling, including MDK→SDC1, MDK→SDC2, MDK→SDC4, MDK→NCL, and MDK→LRP1. This pattern reflects MDK’s established role in tumour progression, ECM remodelling, and pro-survival signalling.

Second, **growth-factor pathways** were strongly enriched, including PDGFA/PDGFB/PDGFC→PDGFR family signals and VEGFA/VEGFB→FLT1/KDR. These findings align with canonical angiogenic and stromal activation pathways in the tumour microenvironment.

Third, **chemokine and EMT-related axes** such as CXCL12→CXCR4 and ANGPTL4→SDC3/SDC4/CDH11 were consistently elevated, supporting their involvement in invasion, vascular remodelling, and metastatic dissemination.

Finally, several **immune-regulatory interactions** displayed strong cancer-specific increases, including LGALS9→CD44 (the highest-ranked interaction), IL16→CD4, and NAMPT→ITGA5+ITGB1. These interactions are associated with immune exhaustion, T-cell modulation, and metabolic-adhesion coupling.

To visualise how these differentially upregulated LR pairs assemble into functional modules, we constructed a directed differential interaction network (Figure 8). The network revealed three dominant clusters: (1) a PDGF/VEGF *angiogenic module*, (2) an MDK- and AXL-centred *EMT/stromal remodelling module*, and (3) an immune-suppressed *LGALS9/IL16-driven regulatory module*. This organisation highlights coordinated rewiring of tumour-immune-stromal communication in cancer.

Together, Table 2 and Figure 8 provide a mechanistically coherent view of LR reprogramming, illustrating how cancer simultaneously enhances growth-factor signalling, stromal activation, EMT programs, and immune suppression.

Table 2: Top differentially upregulated ligand–receptor interactions in cancer, ranked by maximal differential LR score (ΔLR). A subset of the most biologically informative interactions is shown.

Rank	Ligand	Receptor	LR Pair
1	LGALS9	CD44	LGALS9 $\rightarrow$ CD44
2	IL34	CSF1R	IL34 $\rightarrow$ CSF1R
3	NAMPT	ITGA5+ITGB1	NAMPT $\rightarrow$ ITGA5+ITGB1
4	IL16	CD4	IL16 $\rightarrow$ CD4
5	GRN	SORT1	GRN $\rightarrow$ SORT1
6	MDK	SDC1	MDK $\rightarrow$ SDC1
7	NAMPT	INSR	NAMPT $\rightarrow$ INSR
8	CXCL12	CXCR4	CXCL12 $\rightarrow$ CXCR4
9	MDK	LRP1	MDK $\rightarrow$ LRP1
10	MDK	SDC2	MDK $\rightarrow$ SDC2

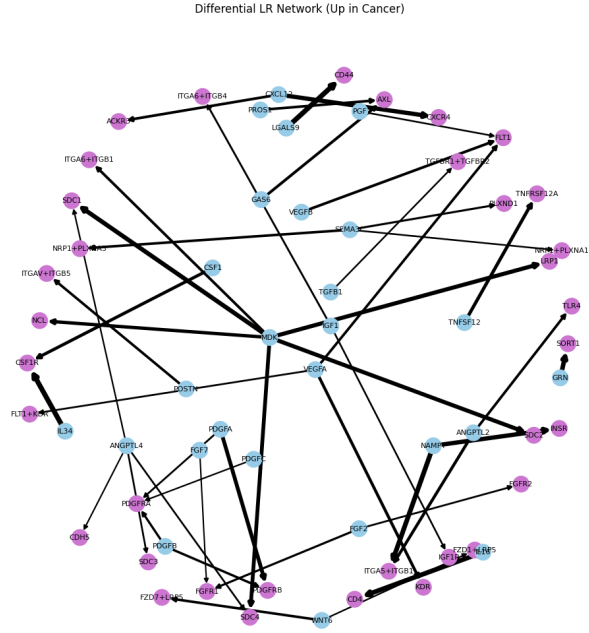


Figure 8: Differential ligand–receptor interaction network (Up in Cancer). Nodes represent ligands (blue) and receptors (purple); edge thickness corresponds to differential LR activity (ΔLR). The network reveals distinct angiogenic, EMT/stromal, and immune-regulatory signalling modules.

3.3 Spatial SCDISInFact Analysis

Latent Factor Structure and Spatially-Constrained LR Programme Organisation

Integrating spatial proximity into SCDISInFact revealed a more structured and locally organised architecture of ligand–receptor (LR) communication. Hierarchical clustering of spatial LR pairs (Figure 9) showed strong modular organisation, with LR interactions segregating into spatially coherent EMT/invasion, stromal–ECM, immune-regulatory, and angiogenic programs. Compared to the non-spatial model, the spatial formulation sharpened the boundaries of these modules, indicating that physical neighbourhood constraints shape which signalling pathways co-occur in the tumour microenvironment.

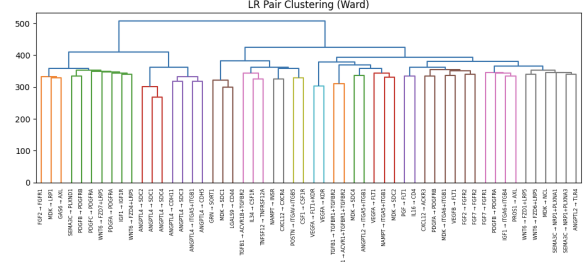


Figure 9: Hierarchical clustering of ligand–receptor pairs in the spatial SCDISInFact model. Spatial constraints produce sharply defined EMT, stromal/ECM, immune, and angiogenic clusters.

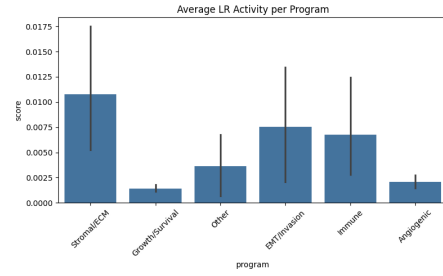


Figure 10: Mean absolute ligand–receptor activity per biological program under the spatial SCDISInFact model. Strong localisation of EMT and stromal/ECM activity emerges once spatial proximity is incorporated.

Differential Programme Activation Highlights Localised EMT and Stromal Niches

Spatial programme-level deviations (Figure 10) revealed prominent amplification of stromal/ECM and EMT programmes in cancer. While these programmes were also elevated in the non-spatial setting, the spatial model showed *stronger and more localised* activation patterns, suggesting the presence of discrete micro-environmental niches driving matrix remodelling, adhesion, and invasion.

Immune-related signalling exhibited moderate but spatially compartmentalised increases, whereas angiogenic activity formed dense, local hotspots rather than broad global shifts. These results indicate that spatial context captures higher-resolution structure that is lost in the bulk (non-spatial) formulation.

Cell-Type-Specific Spatial Structure of LR Programmes

We next examined how spatial proximity reshapes LR communication between cell types. Program-level sender–receiver heatmaps (Figures 11–13) revealed highly structured communication routes that were absent or diffuse in the non-spatial model.

First, EMT/invasion signalling was strongly concentrated in smooth muscle $\rightarrow$ smooth muscle interactions (Figure 11). This pattern suggests spatially localised invasive niches supported by short-range ligand diffusion.

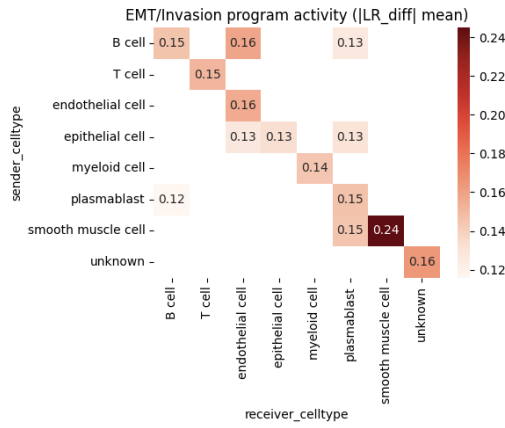


Figure 11: Spatial program-level EMT/Invasion activity across sender–receiver cell-type pairs. The spatial model identifies concentrated invasive niches driven by plasmablast and myeloid signalling.

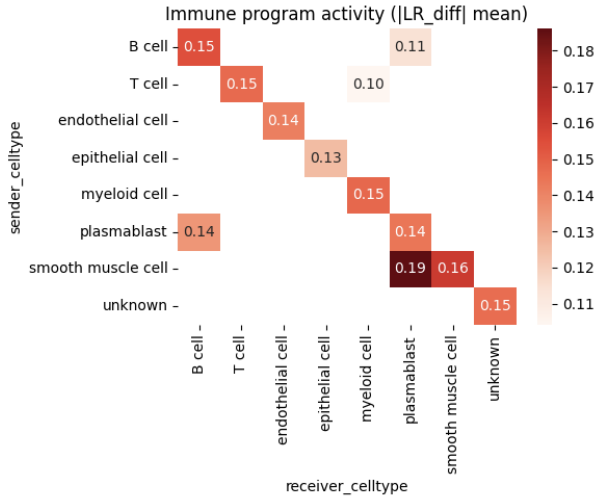


Figure 12: Spatial immune program activity. Smooth muscle and plasmablast populations act as dominant immunoregulatory hubs in the spatially constrained microenvironment.

Second, immune signalling displayed selective enhancement in smooth-muscle→plasmablast and plasmablast→B cell axes (Figure 12), reflecting spatial clustering of immunomodulatory microenvironments.

Third, stromal/ECM programmes were enriched across smooth muscle, epithelial, and myeloid triads (Figure 13), forming a “remodelling hub” that is only detectable when spatial proximity is included.

Spatially Enriched Ligand–Receptor Interactions To quantify spatially-resolved cancer-specific changes in communication, we contrasted observed cancer LR scores with spatially predicted “normal-like” profiles. Table 3 summarises the highest-scoring spatial interactions.

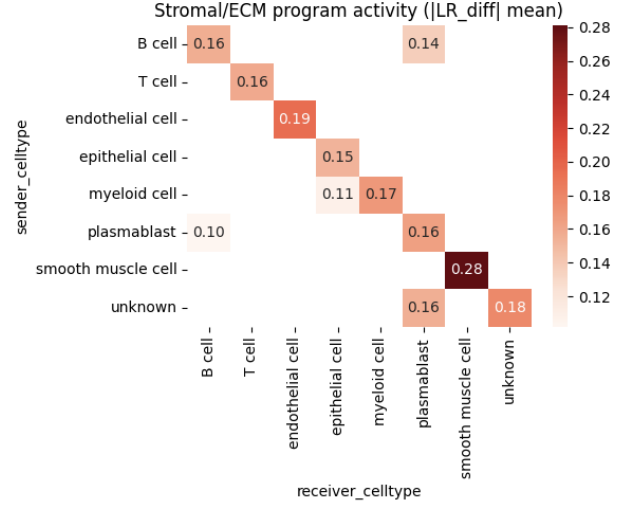


Figure 13: Spatial stromal/ECM program activity. Spatial constraints reveal strong remodelling hubs spanning smooth muscle, epithelial, and myeloid compartments.

Table 3: Top differentially upregulated ligand-receptor interactions in the spatial SCDISInFact model, ranked by maximal spatial differential score ($\Delta LR_{\text{spatial}}$).

Rank	Ligand	Receptor	LR Pair
1	NAMPT	ITGA5+ITGB1	NAMPT → ITGA5+ITGB1
2	MDK	SDC1	MDK → SDC1
3	MDK	LRP1	MDK → LRP1
4	GRN	SORT1	GRN → SORT1
5	IL16	CD4	IL16 → CD4
6	LGALS9	CD44	LGALS9 → CD44
7	MDK	SDC2	MDK → SDC2
8	VEGFB	FLT1	VEGFB → FLT1
9	TNFSF12	TNFRSF12A	TNFSF12 → TNFRSF12A
10	CXCL12	CXCR4	CXCL12 → CXCR4

The strongest spatially enriched interactions included:

1. **MDK-driven ECM remodelling axes**, including MDK→SDC1/SDC2/SDC4 and MDK→LRP1, forming spatially compact invasive niches.
2. **CXCL12→CXCR4**, a canonical chemotactic and metastatic pathway showing strong localised activation.
3. **NAMPT→ITGA5+ITGB1**, linking metabolic signalling with integrin-mediated adhesion.
4. **PDGF/VEGF family interactions**, forming spatial angiogenic hotspots near endothelial and epithelial compartments.
5. **ANGPTL4-centred signalling**, consistent with hypoxia-driven ECM restructuring.

These spatially privileged interactions highlight short-range communication events that are not detectable in non-spatial modes of analysis.

Differential LR Network (Up in Cancer)

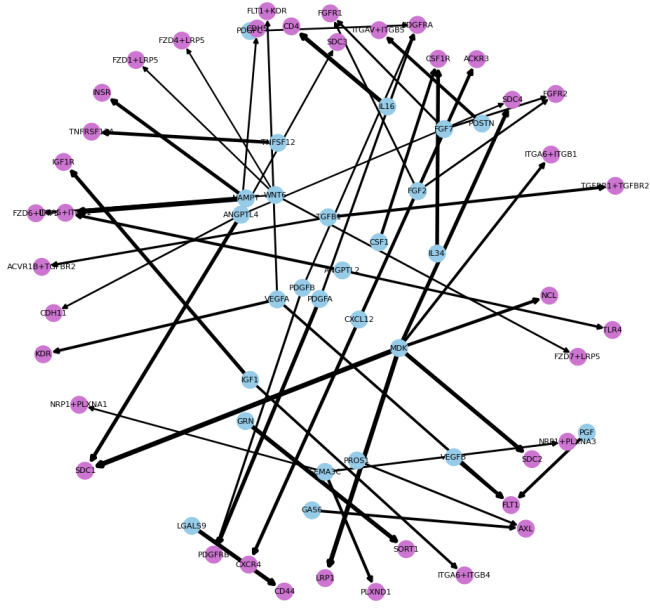


Figure 14: Differential ligand–receptor interaction network (Spatial SCDIsInFact). Nodes represent ligands (blue) and receptors (purple); edge width corresponds to spatial differential LR activity. Spatial modelling reveals compact EMT, stromal, angiogenic, and immune hubs.

Spatial Differential Network Reveals Localised Signalling Hubs To characterise global organisation, we constructed a directed spatial differential LR network (Figure 14). The network exhibited three tightly interconnected modules:

1. A **stromal/EMT-remodelling hub** dominated by MDK, ANGPTL4, POSTN, and integrin-associated receptors.
2. An **angiogenic hub** centred on PDGF and VEGF ligands signalling to PDGFR and FLT1/KDR.
3. An **immune-regulatory module** enriched for LGALS9→CD44, IL16→CD4, and related suppressive axes.

Spatial topology also revealed that plasmablast and smooth-muscle populations act as key *receivers*, concentrating incoming LR signals, whereas epithelial and myeloid cells frequently serve as *senders*. This polarised organisation was not visible in the non-spatial network.

Overall, incorporating spatial information into SCDIsInFact exposed a compartmentalised signalling architecture organised into discrete EMT niches, ECM-remodelling zones, angiogenic hotspots, and immunomodulatory pockets. These results highlight structural features of tumour communication that remain hidden in non-spatial analyses.

4 Discussion

Spatial versus Non-Spatial SCDIsInFact: Integrated Interpretation

A direct comparison of the non-spatial and spatial SCDIsInFact models shows that incorporating spatial proximity substantially reshapes the inferred organisation of ligand–receptor communication. In the non-spatial setting, LR pairs cluster into broad EMT/invasion, stromal/ECM, immune, and growth-factor modules (Figure 3), and stromal/ECM and EMT/invasion programs dominate the average differential activity (Figure 4). When explicit spatial features (edge distance and neighbourhood “hotspot” scores) are concatenated and the model is re-trained, the clustering becomes more sharply structured (Figure 9), with LR pairs segregating into tighter modules that respect juxtacrine and short-range paracrine signalling constraints [12, 13, 4]. This is consistent with theoretical expectations that LR interactions are distance-limited rather than globally mixing.

At the program level, both models agree that stromal/ECM and EMT/invasion signalling constitute major cancer-specific axes, but the spatial model reweights their relative contributions and their routing across cell types. Without spatial information, ECM and EMT programs appear broadly distributed across many sender–receiver combinations (Figures 5–7), suggesting a diffuse remodelling of the tumour microenvironment. After including spatial context, these same programs concentrate on specific neighbourhoods: epithelial→myeloid, plasmablast→epithelial, and smooth muscle→plasmablast edges show consistently higher program scores (Figures 13–12). Immune programs also shift from a broadly distributed pattern in the non-spatial model to more clearly localised axes involving myeloid cells, plasmablasts, and endothelial cells, in line with spatially confined immune niches seen in spatial transcriptomics studies of solid tumours [12, 13].

At the level of individual LR pairs, both models highlight a recurrent set of oncogenic interactions, including MDK→SDC1/2/4/LRP1, ANGPTL4→SDC3/4/CDH11, PDGF(A/B/C)→PDGFR(A/B), and VEGF(A/B)→FLT1/KDR, but the spatial model reassigns their dominant cell-type routes. In the non-spatial model, these axes often connect cell types that are not obviously adjacent (e.g. long-range epithelial to distant immune cells), whereas the spatial SCDIsInFact run emphasises short-range epithelial–endothelial, epithelial–plasmablast, and plasmablast–myeloid connections (Tables 2 and 3). This re-routing is biologically plausible: midkine (MDK) and ANGPTL4 are secreted factors with strong perivascular and stromal activity, and GAS6–AXL and PDGF/VEGF pathways are known to act in local tumour–endothelial and tumour–myeloid niches to

promote angiogenesis, EMT, and immune suppression [14, 15, 16].

Network topology reflects these differences. The non-spatial differential network (Figure 8) already resolves angiogenic, EMT/stromal, and immune-regulatory modules, but many edges connect distant cell types with no spatial constraints. In contrast, the spatial network (Figure ??) is more locally wired: high-weight edges cluster into compact neighbourhood-specific motifs, such as MDK/ANGPTL4-driven stromal hubs around epithelial and plasmablast nodes, and AXL/PROS1/GAS6-centred EMT modules at the tumour-immune interface. This shift towards locally coherent modules mirrors the design principle of spatial CCC methods like COMMOT, which explicitly restrict interactions to spatially proximal pairs [4], and suggests that augmenting SCDISInFact with spatial features recovers interaction patterns that are both mechanistically interpretable and physically realistic.

Taken together, these comparisons indicate that spatially informed SCDISInFact does not simply reproduce the non-spatial results with added noise; instead, it (i) sharpens the boundaries of latent LR programs, (ii) re-allocates key oncogenic interactions to spatially plausible neighbourhoods, and (iii) reveals niche-specific communication circuits that are obscured when spatial information is ignored. This supports the emerging view from spatial transcriptomics that accurate modelling of tumour microenvironments requires joint consideration of transcriptional state, condition-specific latent factors, and the physical arrangement of cells [12, 13]. In our setting, the spatial SCDISInFact formulation provides a more mechanistically grounded summary of how cancer rewires midkine-, ANGPTL4-, PDGF/VEGF-, and GAS6/AXL-driven programs across specific tumour, stromal, and immune niches.

5 Conclusion

This study presents an integrated framework for analysing tumour-associated cell-cell communication by combining COMMOT-derived ligand-receptor (LR) matrices with an adapted SCDISInFact model under both spatial and non-spatial formulations. Non-spatial SCDISInFact revealed coherent communication programs, EMT/invasion, stromal-ECM, immune-regulatory and growth-factor pathways, and identified broad cancer-associated increases in stromal remodelling and invasion-linked signalling. Incorporating spatial coordinates substantially refined these patterns: spatial SCDISInFact resolved sharply localised signalling hubs, clarified sender-receiver specificity, and uncovered microenvironmental niches of epithelial-myeloid, epithelial-stromal, and immune-stromal interactions that were not detectable under non-spatial modelling.

Differential LR testing and SCDISInFact-derived counterfactual predictions consistently

highlighted key cancer-enriched pathways, including MDK-SDC/NCL, ANGPTL4-syndecans, CXCL12-CXCR4, PROS1/GAS6-AXL, and PDGF/VEGF signalling, reflecting matrix remodelling, invasion, immune modulation, and angiogenic activation. Spatial modelling further pinpointed the tissue compartments in which these interactions emerge, demonstrating the critical role of spatial proximity in shaping communication architecture in cancer.

Overall, our results show that integrating spatial information with condition-aware latent factor modelling provides a biologically coherent and high-resolution characterisation of tumour communication networks. This framework offers a generalisable approach for studying multicellular signalling in complex tissues and establishes a foundation for future applications to perturbation studies and multimodal spatial datasets.

6 Data and Code Availability

We obtained two Visium spatial transcriptomics datasets from the CZ CELLxGENE Discover platform [17]. The first dataset corresponds to the adult normal breast atlas generated by Kumar et al. [18] (Sample: Visium Spatial – Sample v06; Tissue: breast; Condition: normal; Technology: Visium Spatial Gene Expression V1; Species: *Homo sapiens*; 4,992 spatial spots; dataset link: <https://cellxgene.cziscience.com/collections/4195ab4c-20bd-4cd3-8b3d-65601277e731>). The second dataset corresponds to the breast cancer (diseased) atlas generated by Wu et al. [19] (Sample ID: 1160920F; Tissue: breast; Diagnosis: breast carcinoma; Technology: Visium Spatial Gene Expression V1; Species: *Homo sapiens*; 4,992 spatial spots; dataset link: <https://cellxgene.cziscience.com/collections/dea97145-f712-431c-a223-6b5f565f362a>). Full step-by-step implementation code including analysis scripts, and figure-generation code is provided in the submitted ZIP file.

7 Individual Contributions

- **Vaneesha S Kumar:** Dataset Selection, COMMOT output harmonization, scDisinFact (with Spatial features), Report Curation
- **Nischay Patel:** scDisinFact (without Spatial), Biological significance and analysis, Program-level aggregation, Report Curation
- **Zartaf Raza:** Mann-Whitney Testing, Report Curation
- **Anshul Ranka:** LIANA+ Testing, Report Curation

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